9. Ascorbic acid, $C_6H_8O_6$, is the main component of vitamin C tablets. Its name is derived from *a-* (meaning "no") and *scorbutus* (scurvy), the disease caused by a deficiency of vitamin C. A student was asked to find the percentage of ascorbic acid in identical vitamin C tablets.

She was told to use the following method.

- Fill a burette with 0.100 mol dm^{-3} sodium hydroxide solution.
- Weigh a conical flask and record its mass.
- Add a vitamin C tablet to the flask, reweigh it and record its mass.
- Add about 50 cm^3 of deionised water to the flask and swirl to break up the tablet.
- Heat the flask gently for 5 to 10 minutes.
- After the solution has cooled add a few drops of a suitable indicator.
- Carry out a rough titration of this solution with the sodium hydroxide solution.
- Accurately repeat the procedure several times and calculate a mean titre.

(a) A **three** decimal place balance was used. The mass of each vitamin C tablet was 500 mg.

Calculate the maximum percentage error in the weighing of the tablet.
You **must** show your working. [2]

Maximum percentage error = ... %

(b) (i) Suggest why she did not need to measure the volume of water accurately. [1]

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(ii) Suggest why she heated the flask for 5 to 10 minutes. [1]

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Examiner only

(c) The student used the results from three titrations to calculate a mean titre. Some of her results are shown below.

Titration	1	2	3
Final reading / cm^3	26.90	26.90	
Initial reading / cm^3	0.25	0.15	0.20
Titre / cm^3	26.65	26.75	

Mean titre = 26.73 cm^3

Determine the **final reading** for the third titration. [2]

Final reading = ... cm^3

(d) Ascorbic acid can decompose upon exposure to air. If this reaction occurred before the titration was completed, state how it might affect the titration results. Explain your answer. [2]

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Examiner only

(e) The equation for the reaction between ascorbic acid and sodium hydroxide is given below.

$$C_6H_8O_6 \quad + \quad NaOH \longrightarrow C_6H_7O_6Na \quad + \quad H_2O$$

M_r 176

The percentage of ascorbic acid is identical in each 500 mg tablet. Calculate the percentage of ascorbic acid in each vitamin C tablet. [3]

Percentage ascorbic acid = ... %

(f) Sulfuric acid and hydrochloric acid are strong acids.

(i) Calculate the pH of a solution of 0.010 mol dm^{-3} sulfuric acid, H_2SO_4. [2]

pH = ...

(ii) When hydrochloric acid is heated with MnO_2 it reacts according to the following equation.

$$MnO_2(s) \quad + \quad 4HCl(aq) \longrightarrow MnCl_2(aq) \quad + \quad Cl_2(g) \quad + \quad 2H_2O(l)$$

Explain why this can be classified as a redox reaction. [2]

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